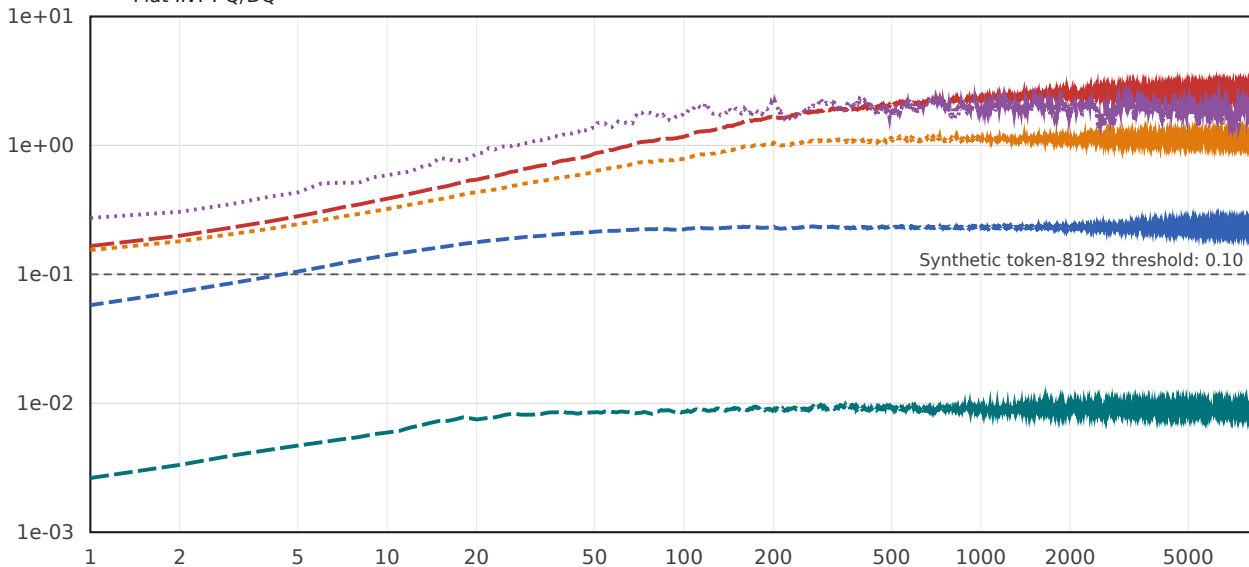


Recurrent-state relative L2 error

- BF16 operands/state (FP32 accum)
- Native encoded MXFP4
- MXFP4 floating Q/DQ (act B32, state B32)
- MXFP4 Q/DQ + MXFP8-E4M3 state (act B32, state B32)
- Flat INT4 Q/DQ



Decode token index (log scale)

Synthetic development nominal; seed=64370; recurrence core; KxV state; 16 Q/K + 32 value heads; K=128, V=128; act/state B32/B32
Single deterministic trace (no uncertainty interval); relative L2 denominator= $\max(|\text{reference}|^2, 1e-12)$; Floating Q/DQ baselines plus native encoded MXFP4